

OpenClaw: The AI Agent That Actually Does Things—and Sometimes Too Much

A Security Analysis of Autonomous AI Agents

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ABSTRACT — This paper examines OpenClaw, an open-source autonomous AI agent capable of executing real-world actions through messaging platforms. While OpenClaw represents significant advancement in personalized AI, its high-privilege access introduces critical vulnerabilities. Through analysis including the Personalized Agent Security Bench (PASB), we evaluate both transformative potential and inherent risks of deploying such agents.

I. Introduction

OpenClaw is a free, open-source autonomous AI agent running on local hardware, interacting via WhatsApp, Telegram, Discord, and Signal. Unlike conventional chatbots, OpenClaw executes real-world actions: sending emails, managing calendars, file operations, shell commands, and browser automation—all triggered by text message.

Released November 2025 by Peter Steinberger as Clawdbot, later Moltbot, now OpenClaw—the name reflects open-source status and lobster-themed branding. Steinberger positions it as "the AI that actually does things," bridging conversational AI and system automation.

"OpenClaw is an AI that actually does things, not just sits there answering questions in a browser."

— P. Steinberger, Creator

II. Architecture

130+

SKILLS

30-70%

ATTACK RATE (NO DEFENSES)

10-30%

ATTACK RATE (WITH DEFENSES)

5+

PLATFORMS

IV. Security Analysis

Early 2026, researchers coined "malevolent AI agent" describing how autonomous systems can be subverted for abuse, exfiltration, and surveillance. Wired and Forbes highlight risks:

- File exfiltration
- Malicious messaging
- Browser manipulation
- Memory poisoning

Attack Vectors (PASB Framework)

- Direct injection:** Malicious instructions
- Indirect injection:** Poisoned content
- Tool deception:** Manipulated API responses
- Memory poisoning:** Malicious context

Model	No Defenses	With Defenses
Llama-3.1-70B	30-70%	10-30%
Qwen-2.5-7B	30-70%	10-30%
GPT-4o-mini	30-70%	10-30%

OpenClaw combines an LLM backbone (Claude, GPT-4, Llama-3, Qwen) with extensible tools for multi-step task planning. Local runtime manages persistence, orchestration, and long-term memory. Runs as persistent daemon on local machine or VPS, maintaining context across sessions.

User → LLM → Plan → Tools → Memory

III. Capabilities

Integration with messaging provides always-available assistance:

- Multi-day inbox triage
- Automated flight booking
- Calendar coordination
- Code review automation

"OpenClaw doesn't just suggest steps; it actually performs them."

"OpenClaw exhibits critical vulnerabilities at user prompt processing, tool usage, and memory retrieval."

— PASB Research, 2026

V. Skills Registry

Over 130 high-privilege operations:

Messaging Financial File R/W Shell OAuth Password Reset
Browser Auto Webhooks Wallet Social Media CI/CD Calendar
Email

VI. Ecosystem

Moltbook, an agent-only social network, enables machine-scale interaction. Agents consuming community content face indirect injection risks.

Ecosystem Risk: Agent-to-agent communication exponentially increases attack surfaces.

VII. Mitigation Strategies

For Developers

- Strict host access control
- Sandboxed environments
- Granular permission scopes
- Comprehensive audit logging
- PASB framework testing

For Users

- Treat as high-privilege system
- Limit account permissions
- Review memory and logs
- Network segmentation
- Offline backups

"OpenClaw is not something you just install and forget; it's a powerful automation tool that can both boost productivity and become a severe security liability if left unattended."

— CrowdStrike, 2026

VIII. Conclusion

OpenClaw has evolved from experiment to symbol of transition from chatbots to acting agents. Local execution, messaging integration, and workflow automation make it compelling for enthusiasts and researchers.

However, it embodies the dual-use dilemma: identical capabilities enabling productivity—file access, communication, system interaction—can be weaponized for exfiltration and manipulation when safeguards fail.

OpenClaw represents AI's trajectory: powerful, personal, and potentially perilous. Robust safeguards are essential.

References

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